Noisy VQE Optimization and Backend-Calibrated Noise

Removing two idealizations from the Block-4 noise study

Seminar notes

July 5, 2026

Abstract

The Block-4 noise study so far made two convenient assumptions: the variational state was optimized on a *noiseless* landscape (noise applied only at evaluation), and the noise itself was a *hand-set uniform* model (one per-`cx` depolarizing rate, or one global T_1/T_2). This note documents lifting both. First, we put the noise *inside* the optimizer, so every cost evaluation is a noisy density-matrix simulation. The finding is a clean dichotomy: for **unital** noise (depolarizing) noise-aware optimization changes nothing – which retroactively *validates* the noiseless-preparation idealization – while for **non-unital** noise (amplitude damping) it raises the *measured* edge string but *lowers* the noiseless fidelity to the true ground state. Minimizing the device-level cost is therefore not the same as preparing the state: a better number can be a worse state. Second, we replace the toy noise with `NoiseModel.from_backend` on a recorded IBM device snapshot (`fake_manila`); the calibrated curve sits *between* the ideal and the harsh uniform toy, i.e. the uniform assumption overstates the damage. Code: `src/block4.py`, plots 8 and 9.

1 Setup: what was idealized

The Block-4 pipeline prepares the even-parity ground state with a parity-penalized VQE cost

$$C(\theta) \;=\; \langle H\rangle_\theta \;+\; \lambda\big(1-\langle \hat{P}\rangle_\theta\big)^2, \tag{1}$$

and then reads the non-local edge string $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1}\rangle$ on the prepared state. Two idealizations entered:

1. **Noiseless preparation.** $C(\theta)$ was evaluated on the pure statevector; the optimizer never saw noise and returned the ideal optimum θ^*. Noise was applied only afterward, to the fixed θ^*.
2. **Toy noise.** The channel was a single uniform per-`cx` depolarizing probability p_{cx}, or one global thermal T_1/T_2 applied identically to every qubit.

Sections 2 and 3 remove these one at a time.

2 Noisy VQE optimization

2.1 Method

We replace the statevector cost by its noisy counterpart: the transpiled preparation circuit is run as a density matrix through a Qiskit `NoiseModel`, and

$$C_{\text{noisy}}(\theta) = \mathrm{Tr}\big[H\,\rho_\theta\big] + \lambda\big(1-\mathrm{Tr}[\hat{P}\,\rho_\theta]\big)^2, \qquad \rho_\theta = \mathcal{E}\big(U(\theta)|0\rangle\langle 0|U^\dagger(\theta)\big), \tag{2}$$

	noisy $\lvert\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1}\rangle\rvert$		noiseless fidelity to ED GS	
strength	best-case	noise-aware	best-case	noise-aware
Depolarizing (unital), p_{cx}:				
0.04	0.711	0.712	0.998	1.000
0.08	0.502	0.501	0.998	1.000
0.12	0.349	0.366	0.998	1.000
Amplitude damping (non-unital), γ:				
0.04	0.656	**0.755**	0.998	0.999
0.08	0.422	**0.565**	0.998	0.994
0.12	0.266	**0.337**	0.998	**0.966**

Table 1: Best-case (θ^*, noiseless optimum) vs. noise-aware (re-optimized under the channel), $L = 4$, $r = 3$, $\mu = 0$, $\lambda = 0.1$, seed 7. Under depolarizing noise both quantities are unchanged. Under amplitude damping the measured edge string *rises* while the noiseless fidelity to the true ground state *falls* at strong damping.

where $\mathcal{E}$ fires a channel after every `cx`. Each evaluation is one 4^L density-matrix simulation, so the optimizer (COBYLA) is warm-started from the noiseless optimum θ^* and refined, with a few random restarts; the best noisy cost is kept. The warm start guarantees the two strategies coincide at zero noise. Everything is at $L = 4$, $r = 3$, $\mu = 0$ (sweet spot), $\lambda = 0.1$.

2.2 Unital vs. non-unital noise

The two columns of Figure 1 sweep the *same* per-`cx` error axis with two channels of opposite character:

- **Depolarizing** is *unital*, $\mathcal{E}(I) = I$: it contracts all traceless expectations toward the maximally mixed state without singling out a direction. Empirically the noise-aware optimum is indistinguishable from θ^* in every observable (Table 1, top): there is nothing for re-optimization to exploit.

- **Amplitude damping** is *non-unital*: it drives every qubit toward $|0\rangle$, breaking the symmetry. The energy optimum genuinely moves, and the noisy optimizer finds a different θ.

2.3 The fidelity honesty check

A lower noisy energy or a higher measured edge string is meaningless if the underlying state has drifted *away* from the true Majorana ground state – the optimizer would then be exploiting the channel rather than preparing the state better. The decisive quantity is therefore the **noiseless** fidelity $|\langle\psi_{\mathrm{ED}}|\psi(\theta)\rangle|^2$ of the re-optimized parameters (Table 1, last columns; Figure 1, bottom row).

The result is unambiguous. At *weak* damping ($\gamma = 0.04$) the noise-aware state keeps essentially the same fidelity ($0.998 \to 0.999$) while raising the edge string – genuine variational compensation. At *strong* damping ($\gamma \gtrsim 0.06$) the edge string still rises, but the noiseless fidelity *drops* (to 0.94–0.97) and does so non-monotonically and seed-dependently: the “recovery” is partly the optimizer gaming the observable through the channel, not a better-prepared ground state. So the honest headline is not “noise-aware VQE recovers the signal” but:

minimizing the device-level cost is not the same as preparing the ground state.

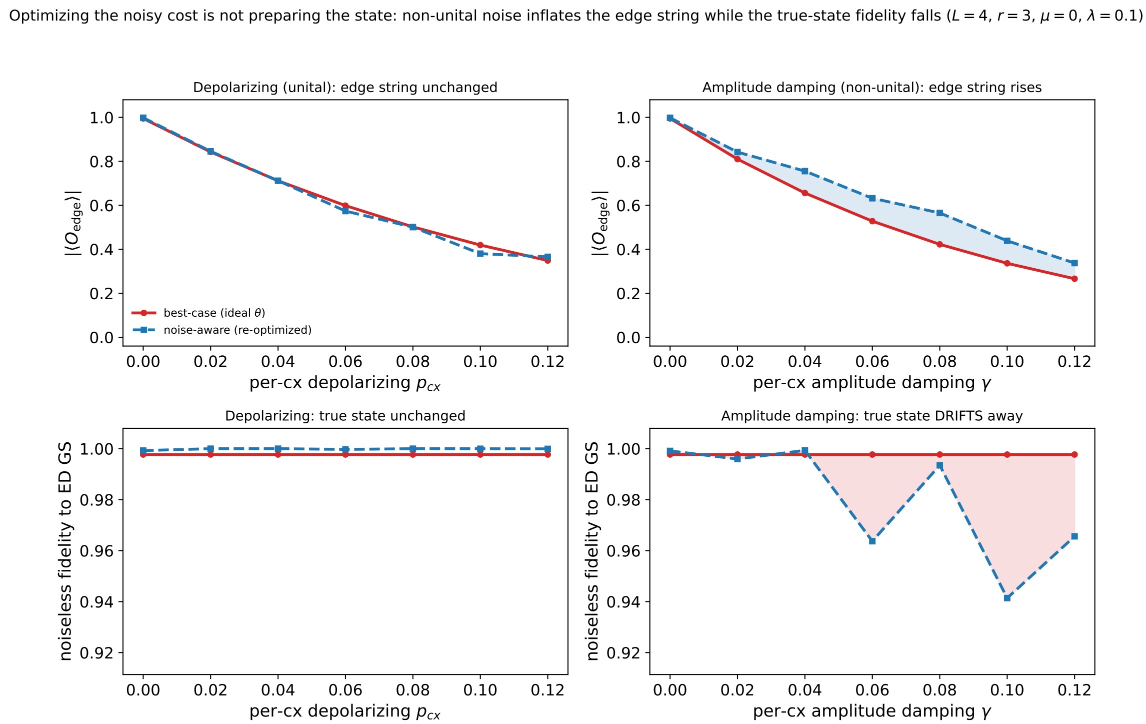

Figure 1: Best-case vs. noise-aware optimization. Columns: depolarizing (unital) | amplitude damping (non-unital), same per-`cx` axis. Top: the *measured* noisy edge string; bottom: the *noiseless* fidelity to the ED ground state. Non-unital noise lets the optimizer inflate the top-row observable while the bottom-row fidelity falls.

Conversely, the unital null is itself a useful result: it shows that for symmetric gate noise the noiseless-preparation idealization of the rest of Block 4 loses nothing – the best-case curves are representative, not optimistic.

3 Backend-calibrated noise

3.1 Method

`NoiseModel.from_backend(backend)` builds a channel from a real device's recorded calibration – per-qubit T_1/T_2, per-*gate* error rates, gate durations, and readout error – rather than one hand-set number. We use the IBM `fake_manila` snapshot (a five-qubit linear device shipped with `qiskit_ibm_runtime`); its `cx` error rates are heterogeneous, $\approx$ 0.009–0.014, and its linear coupling matches the chain, so every nearest-neighbour `cx` of the ansatz carries a calibrated error. The fixed VQE angles are then evaluated under this model and compared with the uniform toy on the identical circuit.

3.2 Result

At the sweet spot the calibrated model costs a meaningful $\sim$12% of the edge signal, but far less than the deliberately harsh uniform toy: because the real per-`cx` rates (~ 0.01) are roughly five times smaller than the toy's 0.05, a uniform toy model *overstates* the damage (Table 2, Figure 2). The

$\lvert\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1}\rangle\rvert$ at $\mu = 0$	ideal	backend (`fake_manila`)	uniform toy ($p_{cx} = 0.05$)
value	0.99	0.87	0.67

Table 2: The calibrated device curve sits between the ideal and the uniform toy.

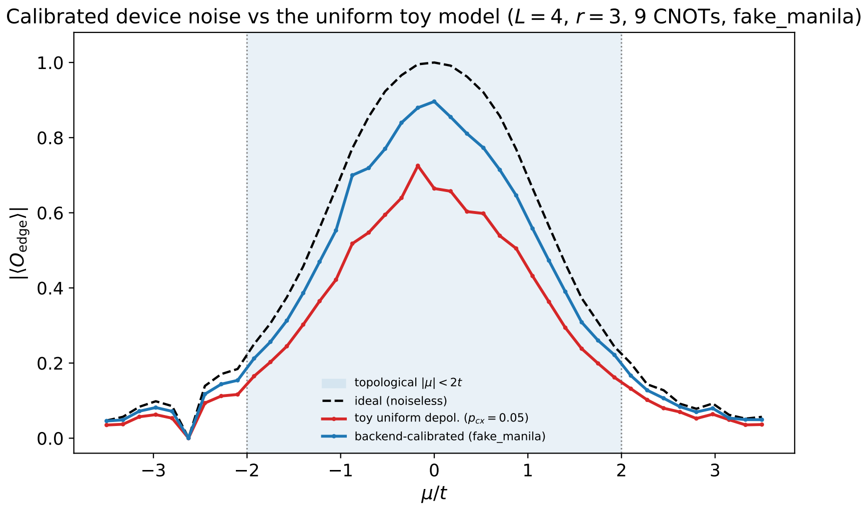

Figure 2: Edge string across the phase diagram under ideal, uniform-toy, and backend-calibrated (`fake_manila`) noise. The calibrated curve lies between the other two.

dominant driver is the per-gate error *magnitude*; per-qubit heterogeneity is a secondary refinement at $L = 4$.

3.3 Caveats

- **Readout error is excluded.** The diagnostic is read as $\mathrm{Tr}[X_0 Z_1 \cdots Z_{L-2} X_{L-1}\, \rho]$ directly from the saved density matrix, with no measurement instruction in the circuit. `from_backend` does include a `measure` error, but it never fires on this path – so even the device curve is mildly optimistic, since real readout adds 1–3% per qubit.
- **Snapshot, not live hardware.** `fake_manila` is recorded calibration data, representative of the device at snapshot time, not a live run. It requires `qiskit_ibm_runtime`; absent that, the code falls back to a synthetic `GenericBackendV2`, whose (seed-generated) error rates are much smaller and should *not* be read as a real-device result.

4 Takeaways

- Optimizing a noisy cost does not by itself buy a better *state*. For symmetric (unital) gate noise it changes nothing; for non-unital noise it can raise a chosen observable while degrading fidelity to the target – an artifact to be reported honestly, not a mitigation.
- The unital null result is the quiet payoff: it certifies that the noiseless-preparation idealization used elsewhere in Block 4 is not hiding an optimistic bias for depolarizing noise.

- Uniform toy noise is a conservative caricature: a calibrated device model is milder, though still not noise-immune, and the device curve here is itself optimistic because readout error is excluded.

Reproduce.

```
python src/block4.py --plots 8   # noisy VQE optimization (unital vs non-unital)
python src/block4.py --plots 9   # toy vs backend-calibrated noise
```

Plot 8 uses `amp_damp_noise_model` / `noisy_optimize`; plot 9 uses `backend_noise_model` (`NoiseModel.from_backend`).